MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10545702

(Ballistics Data Amplifier)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly hereinafter referred to as assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Ballistic Data Amplifier Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50785	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

10545425	Sight Current Generator
10545455	Receptacle
10545702	Component Assembly, Ballistic Data Amplifier

PACKAGING DATA SHEET

10545702	Packaging of Circuit Card Assembly, Ballistic Data Amplifier

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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REQUIREWNTS

3.1 item definition. This assembly is ● single-ended printed circuit board consisting of four operational amplifiers. The circuit card is used to sum, invert, *and/or* amplify the multiple inputs which make up the magnet and torque current voltages developed in the Sight Current Generator Drawing Number 10545425 in which the circuit card is assembled.

3.1.1 Major component list. The components *of this* assembly shall be as specified on the Engineering Parts List (EPL) 10545702.

3.1.2 Government furnished property. Not applicable.

3.2 Characteristics

3.2.1 *Performance.* *The* ● ssembly shall be capable *of performing* the following requirements when assembled in ● accordance with Drawing N-or *10545702.*

8, Operational amplifier A1 shall sum the inputs required to develop the torque current voltage.

b. Operational *amplifier A2* shall *amplify* and invert the input required *to* drive the self test circuit.

c. operational amplifier A3 shall sum the inputs required to develop the magnet current voltage.

d. Operational amplifier A4 shall invert *one of* the inputs required to develop the magnet current voltage.

3.2.1.1 Electrical requirements. The *offset* voltage *of each* operational amplifier in the assembly shall be *zero VDC* ± *SO* millivolts with zero VDC ●pplied to the I n p u t The input and output pins shall be as specified in Table 1.

Table I. Input and Output Pins

Amlifiers	Input Pins	Output Pins
Al	.2, 3, 9, 10	5
A2	15	
A3	8, 19, 20, 21, 22, 23, 26	12
A4	8	16

3.2.1.2 Input and output levels

3.2.1.2.1 Input power shall be supplied to the Ballistic Data Amplifier8a specified in Table II.

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Table II. Input POWer

Pin	Volt age
4, 18, 25	+15 Vdc ± 2. 0 percent
14	-IS Vdc ± 2.O percent
1, 28	Ground

3.2.1.2.2 Output levels. TM requirements specified in 3.2.1.2.2.1 through 3.2.1.2.2.4 shall be met when supplied with the voltage input specified (Vin) in Table II. Vo • Vin (K)

3.2.1.2.2.1 Pin S output The output between pin 5 and pin one shall be as specified in a, b, c, and d.

a. Vin (-.2764) ± 1.5 percent with Vin of -13 Vdc to +13 Vdc between pin 2 and pin one, when pins 3, 9 and 10 are at zero Vdc.

b. Vin (-4.802) 21.0 percent with Vin of -1.7Vdc to +1.7 Vdc between pin 3 and pin one, when pins 2, 9 and 10 are at zero Vdc.

c. Vin [+.0508) ± 15 percent with Vin of -13 Vdc to +13 Vdc between pin 9 and pin one, when pins 2, 3 and 10 are at zero Vdc.

d. Vin (-.6317) ± 1.0 percent with Vin of -13 Vdc to +13 Vdc between pin 10 and pin one, when pins 2, 3 and 9 are at zero Vdc.

3.2.1.2.2.2 Pin 12 output: The output shall be Vin (-7.634) ± 1.0 percent with Vin of -1.7 Vdc to zero Vdc between pin 15 and pin 28 and zero + 0.50 Vdc with Vin of zero to +1.7 Vdc between pin 15 and pin 28.

3.2.1.2.2.3 Pin 16 output: The output shall be Vin (-1.-) ± 0.5 percent With Vin of -13 Vdc to +13 Vdc between pin 8 and pin one.

3.2.1.2.2.4 Pin.17 output: The output between pin 17 and pin 28 shall be as specified in a through g.

a. Vin (+.1599) ± 1.0 percent with Vin of -4 to +13 Vdc between pin 8 and pin 28 and -1.080 Vdc ± 1.0 percent with Vin of -6 to -13 Vdc between pin 8 and pin one when pins 19, 20, 21, 22, 23 and 26 are at zero Vdc.

b. Vin (-.0331) 22.0 percent with Vin of -13 Vdc to +13 Vdc between pin 19 and pin 28 when pins 8, 20, 21, 22, 23 and 26 are at zero Vdc.

c. Vin (-.0309) ± 2.0 percent with Vin of -13 Vdc to +13 Vdc between pin 20 and pin 28 when pins 8, 19, 21, 22, 23 and 26 are at zero Vdc.

d. Vin (-.3434) ± 1.0 percent with Vin of -13 Vdc to +13 Vdc between pin 21 and pin 28 when pins 8, 19, 20, 22, 23 and 26 are at zero Vdc.

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e. Vin (-.8768 tO. S percent with Vin of -13 Vdc to +13 Vdc between pin 22 and pin one, when pins 8, 19, 20, 21, 23 and 26 are at zero Vdc.

f. VIN (-.0816)} ± percent with Vin*-lSVdcto+lS Vdc between pin 23 and-pin 28, when pins 8, 19, 20, 21, 22 and 26 are. ● at zero Vdc.

g, Vin (-.84Ol) ± 0.5 percent with Vin of -13VdC to +l3 VdC between pin 26 and pin 28, when pins 16, 19, 20, 21, 22 and 23 are ● t zero vdC.

3.2.1.3 Environmental. The assembly shall most the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristic. The characteristics shall be as specified in drawing 10545702 and the engineering puts list.

3.3 Design and construction

3.3.1 Production drawings. This assembly shall b. fabricated in accordmce with the requirments of specification MIL-C-50785, drawing 10545702 and ● ssociated lists.

3.3.2 Standards of Manufacture. The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship. The workmanship shall be as specified in *specification* MIL-C-50785

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4. QUALITY ASSURANCE PROVISION

4.1 General.

4.1.1 Responsibility for inspection. The responsibility for inspection shall be as specified in specification MIL-C-5078S.

4.1.2 Special tests and examination. The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections. The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-5078S.

4.2.1 Test equipment. (See Figure 1).

4.2.1.1 Power supply PS1). The power supply shall be 15 50.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (M V) or 100 mv peak to peak.

4,2.1.2 Power surely (PS2). The power supply shall be -15 ± vdc, 100 ma or greater, ripple less than 10 mv ms or 100 mv peak to peak,

4.2.1.3 Powers The power supply shall b. zero ± 13 Vdc, with adjustment an regulation to ± .01 percent, 10 ma or greater.

4.2.1.4 Resistors. Two resistors are required with a ratings of 1000 Ohms ± 5 percent

4.2.1.5 Digital voltmeter (DVM) .-The Digital Voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

4.2.2 Functional test

4.2.2.1 Test connections: Power supplies PS1 and PS2 and resistors shall be connected to the assembly under test in accordance with Figure 1 using receptacle 10545455.

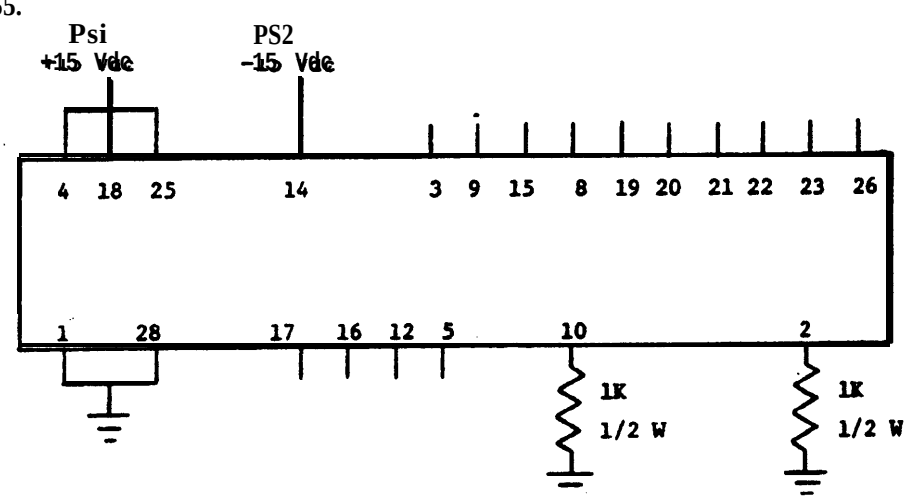


Figure I. Test Connections

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4.2.2.2 Test procedure: Compliance with the requirement *of* 3.2.1.1 through 3.2.1.1.2.2.4 shall be verified as follows:

● . Power supply (MS) voltage shall be adjusted and connected to the specified input pin per Table III. A digital voltmeter shall be used to measure the voltage ●t the ●pplicable output pin to. the limits indicated in Table 111.

b. A digital voltmeter shall read 0.050 Vdc on any one of output pins S, 12, 16, end 17 when pins 2, 3, 8, 9, 10, 15, 19, 20, 21, 22, 23 and 26 are open circuited.

Table III. Test Limits

Input Voltage (Vdc)	Input Pin	Output Voltage (Vdc)	Output Pin
+2.7 ±0.006	3	-12.965 ±0.130	5
+1.7 ±0.006	15	0 ±0.050	12
+13.0 ±0.013	2	-3.593 ±0.054	5
+13.0 ±0.013	9	+0.660 ±0.100	5
+13.0 ±0.013	10	-8.212 ±0.082	5
+13.0 ±0.013	8	-13.000 ±0.065	16
+13.0 ±0.013	8	+2.078 ±0.021	17
+13.0 ±0.013	19	-0.430 ±0.010	17
+13.0 ±0.013	20	-0.401 ±0.010	17
+13.0 ±0.013	21	-4.465 ±0.045	17
+13.0 ±0.013	22	-11.399 ±0.057	17
+13.0 ±0.013	23	-1.061 ±0.011	17
+13.0 ±0.013	26	-10.921 ±0.055	17
-2.7 ±0.006	3	+12.965 ±0.130	5
-1.7 ±0.006	15	+12.976 ±0.130	12
-4.0 ±0.006	8	-0.640 ±0.010	17
-6.0 ±0.006	8	-1.080 ±0.011	17
-13.0 ±0.013	2	+3.593 ±0.054	5
-13.0 ±0.013	9	-0.660 ±0.100	5
-13.0 ±0.013	10	+8.212 ±0.082	5
-13.0 ±0.013	8	+13.000 ±0.065	16
-13.0 ±0.013	8	-1.080 ±0.011	17
-13.0 ±0.013	19	+0.430 ±0.010	17
-13.0 ±0.013	20	+0.401 ±0.010	17
-13.0 ±0.013	21	+4.465 ±0.045	17
-13.0 ±0.013	22	+11.399 ±0.057	17
-13.0 ±0.013	23	+1.061 ±0.011	17
-13.0 ±0.013	26	+10.921 ±0.055	17

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5. PREPARATION FOR DELIVERY

5.1 Packaging Packaging and marking shall be in accordance with the level A requ rements of Packaging Data Sheet [PDS), AMC form 1029, 10545702.

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 • ro ● pplicable to this specification.

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Preparing Activity:

Army-MU

Project No. 1285-A312

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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